

Barrier PDLP

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Barrier PDHG (BPDHG) I

Given the standard-form LP $\min_{\mathbf{x} \in \mathbb{R}^n} \mathbf{c}^\top \mathbf{x}$ s.t. $\mathbf{Ax} = \mathbf{b}$, $\mathbf{x} \geq 0$, the classical PDHG iteration is

$$\begin{aligned}\mathbf{x}^{k+1} &= \arg \min_{\mathbf{x} \geq 0} \left\{ \mathbf{c}^\top \mathbf{x} - (\mathbf{y}^k)^\top \mathbf{Ax} + \frac{1}{2\tau} \|\mathbf{x} - \mathbf{x}^k\|_2^2 \right\} = \max \left\{ \mathbf{x}^k + \tau (\mathbf{A}^\top \mathbf{y}^k - \mathbf{c}), \mathbf{0} \right\}, \\ \mathbf{y}^{k+1} &= \arg \min_{\mathbf{y} \in \mathbb{R}^m} \left\{ -\mathbf{b}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{A} (2\mathbf{x}^{k+1} - \mathbf{x}^k) + \frac{1}{2\sigma} \|\mathbf{y} - \mathbf{y}^k\|_2^2 \right\} = \mathbf{y}^k - \sigma [\mathbf{A} (2\mathbf{x}^{k+1} - \mathbf{x}^k) - \mathbf{b}].\end{aligned}\tag{1}$$

BPDLP adds a log barrier to handle the nonnegativity constraint and replaces the LP with

$$\min_{\mathbf{x} \in \mathbb{R}_+^n} \mathbf{c}^\top \mathbf{x} - \mu \sum_{i=1}^n \log(x_i) \quad \text{s.t.} \quad \mathbf{Ax} = \mathbf{b}.\tag{2}$$

Given its Lagrangian $\mathcal{L}_\mu(\mathbf{x}, \mathbf{y}) = \mathbf{c}^\top \mathbf{x} - \mu \sum_{i=1}^n \log(x_i) - \mathbf{y}^\top (\mathbf{Ax} - \mathbf{b})$, the barrier problem equation 2 is equivalent to the saddle-point problem $\min_{\mathbf{x} > 0} \max_{\mathbf{y} \in \mathbb{R}^m} \mathcal{L}_\mu(\mathbf{x}, \mathbf{y})$. Applying PDHG to this saddle-point problem gives the BPDHG iteration:

$$\begin{aligned}\mathbf{x}^{k+1} &= \arg \min_{\mathbf{x} > 0} \left\{ \mathbf{c}^\top \mathbf{x} - \mu \sum_{i=1}^n \log(x_i) - (\mathbf{y}^k)^\top \mathbf{Ax} + \frac{1}{2\tau} \|\mathbf{x} - \mathbf{x}^k\|_2^2 \right\} = \text{prox}_{-\tau\mu \sum_{i=1}^n \log(x_i)} \left(\mathbf{x}^k + \tau (\mathbf{A}^\top \mathbf{y}^k - \mathbf{c}) \right), \\ \mathbf{y}^{k+1} &= \arg \min_{\mathbf{y} \in \mathbb{R}^m} \left\{ -\mathbf{b}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{A} (2\mathbf{x}^{k+1} - \mathbf{x}^k) + \frac{1}{2\sigma} \|\mathbf{y} - \mathbf{y}^k\|_2^2 \right\} = \mathbf{y}^k - \sigma (\mathbf{A} (2\mathbf{x}^{k+1} - \mathbf{x}^k) - \mathbf{b}).\end{aligned}\tag{3}$$

Barrier PDHG (BPDHG) II

The update of $\mathbf{x}$ can be computed in closed form as (elementwise for each coordinate i):

$$\mathbf{x}^{k+1} = \frac{\mathbf{z}^k + \sqrt{(\mathbf{z}^k)^2 + 4\tau\mu}}{2}, \quad \text{where } \mathbf{z}^k = \mathbf{x}^k + \tau \left(\mathbf{A}^\top \mathbf{y}^k - \mathbf{c} \right). \quad (4)$$

For the original LP, the KKT conditions are

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{A}^\top \mathbf{y} + \mathbf{s} = \mathbf{c}, \quad \mathbf{x} \geq 0, \quad \mathbf{s} \geq 0, \quad \underbrace{x_i s_i = 0, \quad i = 1, \dots, n.}_{\Leftrightarrow \mathbf{Xs} = \text{Diag}(\mathbf{x})\mathbf{s} = \mathbf{0}} \quad (5)$$

The barrier problem has the following μ -perturbed KKT system:

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{A}^\top \mathbf{y} + \tilde{\mathbf{s}} = \mathbf{c}, \quad \mathbf{x} > 0, \quad \tilde{\mathbf{s}} > 0, \quad \underbrace{x_i \tilde{s}_i = \mu, \quad i = 1, \dots, n.}_{\Leftrightarrow \mathbf{X}\tilde{\mathbf{s}} = \mu \mathbf{1}} \quad (6)$$

- Thus, as long as $\mu \downarrow 0$, the Barrier KKT $\rightarrow$ LP KKT, and so will the solutions.
- At the exact solution, the primal-dual gap is $\mathbf{c}^\top \mathbf{x} - \mathbf{b}^\top \mathbf{y} = n\mu$, which determines the optimality scale.
- This links BPDHG to the interior point method (IPM). IPM solves the perturbed KKT system by Newton's method, whereas BPDHG uses PDHG for a fixed μ . Since the equality constraint $\mathbf{Ax} = \mathbf{b}$ is unchanged, the $\mathbf{y}$ update is also unchanged.

From BPDHG to Barrier PDLP I

Recall, in short, the BPDHG given μ procedure: (1) $\mathbf{z}^k = \mathbf{x}^k + \tau(\mathbf{A}^\top \mathbf{y}^k - \mathbf{c})$, (2) $\mathbf{x}_i^{k+1} = \frac{\mathbf{z}_i^k + \sqrt{(\mathbf{z}_i^k)^2 + 4\tau\mu}}{2}$, (3) $\mathbf{y}^{k+1} = \mathbf{y}^k - \sigma(\mathbf{A}(2\mathbf{x}^{k+1} - \mathbf{x}^k) - \mathbf{b})$.

Algorithm 1: BPDLP with barrier stages and adaptive restarts (Algorithm 2 in the paper)

Input: initial point $(\mathbf{x}^0, \mathbf{y}^0)$, barrier parameter μ_0 , and dual-primal weight ω_0

Initialize: $j \leftarrow 0$ and $n \leftarrow 0$

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1 while the original LP is not solved do
2     Fix  $\mu \leftarrow \mu_j$  // barrier stage  $j$ 
3     while barrier stage  $j$  is not complete do
4         Reset epoch statistics and set  $t \leftarrow 0$  // restart epoch  $n$ 
5         while restart is not triggered do
6             Choose the adaptive step  $\eta_t$  // inner iteration  $t$ 
7             Compute primal-dual stepsize  $(\tau_t, \sigma_t)$  from  $(\eta_t, \omega_n)$  by  $\tau_t = \eta_t / \omega_n$  and  $\sigma_t = \omega_n \cdot \eta_t$ 
8             Perform one BPDHG step
9             Update the weighted average  $(\bar{\mathbf{x}}^t, \bar{\mathbf{y}}^t)$ 
10            Test the current and averaged points; check LP stopping
11            Choose a restart point and update  $\omega_n$ 
12             $n \leftarrow n + 1$ 
13        Update  $\mu_{j+1}$  and optionally predict the next starting point
14     $j \leftarrow j + 1$ 
```

Notes on some implementation details:

- The restart loop is inherited from PDLP. For each fixed μ_j and restart epoch n , the algorithm maintains a weighted average of the iterates $\bar{\mathbf{z}}^{n,t} := (\bar{\mathbf{x}}^t, \bar{\mathbf{y}}^t)$ using weights η_t . It then compares the last iterate $\mathbf{z}^{n,t}$ with the weighted average $\bar{\mathbf{z}}^t$ and selects the better point to initialize the next restart epoch.
- The direct stepsizes τ_t and σ_t are controlled by η_t and ω_t . The parameter η_t determines the overall stepsize of the current BPDHG step and is chosen adaptively to remain reasonably large.
- Warm start: when a barrier stage terminates normally and shifts from μ_j to μ_{j+1} , the algorithm predicts improved starting points $\mathbf{x}^{j+1,0}$ and $\mathbf{y}^{j+1,0}$ for the next barrier stage.

The update of μ_j is as follows.

- If the μ_j -KKT system is solved sufficiently well, apply the larger reduction $\mu_{j+1} = \theta \mu_j$ (e.g. $\theta = 0.2$). Here, “solved sufficiently well” means:
 - KKT residual $\text{KKT}_{\mu_j} \leq \max\{C\mu_j, \vartheta \text{KKT}_{\text{LP}}\}$ (e.g. C not specified, $\vartheta = 0.05$).
 - KKT residual includes: $\mathbf{r}_{\text{prim}} = \mathbf{Ax} - \mathbf{b}$, $\mathbf{r}_{\text{dual}} = \mathbf{c} - \mathbf{A}^\top \mathbf{y} - \mu_j \mathbf{x}^{-1} \mathbf{e}$. Thus, basically it's something like $\text{KKT}_{\mu_j} \approx \max\{\|\mathbf{r}_{\text{prim}}\|, \|\mathbf{r}_{\text{dual}}\|\}$.
 - Similar to KKT-LP, $\text{KKT}_{\mu_j} \approx \max\left\{\frac{\|\mathbf{Ax} - \mathbf{b}\|}{1 + \|\mathbf{b}\|}, \frac{\|[\mathbf{A}^\top \mathbf{y} - \mathbf{c}]_+\|}{1 + \|\mathbf{c}\|}, \frac{|\mathbf{c}^\top \mathbf{x} - \mathbf{b}^\top \mathbf{y}|}{1 + |\mathbf{c}^\top \mathbf{x}| + |\mathbf{b}^\top \mathbf{y}|}\right\}$.
- If the μ_j -KKT system is not solved sufficiently well before the inner budget is exhausted, apply the slower reduction $\mu_{j+1} = \tau_{\text{inner}} \mu_j$ (e.g. $\tau_{\text{inner}} = 0.9$).

References I

- [1] Yingxin Zhou, Stefano Cipolla, and Phan T. Vuong. A barrier primal dual hybrid gradient method for solving linear programming problems, 2026. URL <https://arxiv.org/abs/2608.26667>.